

Did Generative AI Trigger a Structural Break?

ARIMA Analysis of XLK ETF

1 Introduction

Artificial Intelligence (AI) has been a hot topic in recent years, with its wide applications in scientific discovery, advanced cybersecurity, and personalized AI agents. In the stock market, technology stocks have accounted for 32% of global equity returns and 40% of total U.S. equity market returns since 2010 (Goldman Sachs, 2024), playing a dominant role in modern financial markets. Large-cap firms, such as Apple and Microsoft, represent a substantial share of total market capitalization; consequently, their performance increasingly shapes the trajectory of major stock indices, including the S&P 500 and the NASDAQ Composite (Yahoo Finance). Since Technology companies are largely valued based on expectations of future innovation and scalability, breakthroughs can trigger significant revisions in investor sentiment, leading to movement in stock prices.

Breakthrough in new technology also impacts a firm's business model. For technology-centric companies in particular, launching a novel product often increases company's net valuation (McKinsey, 2016). As several tech giants have launched flagship AI products, such as Gemini from Google and Claude from Anthropic, their market valuations and stock prices have shifted accordingly (Investing.com, 2026). The impact of novel AI product is seemingly significant since it can impact the company's valuation that the company does not even need to be publicly traded. The hard launch of ChatGPT in November 2022 significantly increased OpenAI's, a non-profit artificial intelligence research company, private valuation and attracted substantial investment, with its estimated worth skyrocketing from roughly \$20 billion to \$85.2 billion by 2026 (Barrientos, 2026).

Furthermore, the adoption of AI applications has also reshaped market narratives surrounding technological growth and productivity. Beyond the firms that directly launch AI models, companies perceived as key beneficiaries of AI infrastructure, most notably the GPU manufacturing company NVIDIA Corporation, have also experienced significant revaluations (Morningstar, 2026). This shift suggests that innovation shocks may influence not only short-term prices but also the broader structural dynamics of the overall technology sector.

With the crucial role of technology stock in the general financial market and AI product's impacts on company's evaluation, we want to investigate whether the technological breakthrough impact the technology sector stock market. This report investigates:

- Did the stock return pattern changes after the emergence of AI products?
- What's the predicted stock return based on current stock market trend?

Specifically, we evaluate this change by analyzing the Technology Select Sector SPDR Fund (XLK) using ARIMA models to examine whether the release of ChatGPT corresponds to a statistically significant shift in XLK price dynamics from 2017 to 2024.

Understanding whether such a technological breakthrough corresponds to a lasting change in market behavior is crucial for investors and policymakers. Persistent shifts in return pattern can affect risk assessment, portfolio allocation, and forecasting models in financial market analysis. Financial return series are time-dependent and often exhibit changes in volatility and growth dynamics. If the emergence of artificial intelligence has altered investor expectations, this effect may manifest as a structural break in the return-generating process.

2 Methodology

2.1 Data Overview

This report uses the daily XLK fund return price from Investing.com, a financial data platform that offers real-time data. The XLK stock price is collected by S&P Dow Jones Indices and State Street Global Advisory. S&P Dow Jones Indices is in charge of creating the company benchmark on entering the technology company stock pool. The State Street Global Advisors selected particular stocks into the combination of the XLK stock.

We chose the XLK because it's a comprehensive fund that captures an effective representation of the technology sector of the S&P 500 Index. The XLK has a 4-star in the Morningstar rating by March 2026, indicating an above-average performance that the XLK is in the top 22.5% to 32.5% of technology category. The XLK ETF provides precise exposure to companies from technology hardware, storage and peripherals; software; communications equipment; semiconductors and semiconductor equipment; IT services; and electronic equipment, instruments and components industries (State Street). This feature allows investors to take strategic or tactical positions at a more targeted level than traditional style based investing. The companies in the XLK are selected from the S&P 500 based on the Global Industry Classification Standard (GICS) where companies are assigned to a single classification based on revenue.

2.2 Data Collection

Using the dataset from the investing website, we derive the following variables for the analysis:

- **Price** (in USD): The final price at which the ETF traded when the stock exchange closed (typically 4:00 PM EST).
- **Open** (in USD): The price of the very first transaction when the market opened (9:30 AM EST).
- **High & Low** (in USD): The maximum and minimum prices reached during the trading session.
- **Vol.**: The total number of shares traded during the day (e.g., 15.90M means 15.9 million shares).
- **Change**: The percentage difference between the current day's closing price and the previous day's closing price.

The dataset comprises daily observations spanning from January 3, 2017, to March 28, 2026. This time frame was specifically selected to ensure that the launch of ChatGPT (November 30, 2022) serves as a central pivot point, providing a balanced sample size for comparing pre- and post-innovation market dynamics.

2.3 Data Cleaning

In the raw dataset from the investing website, we observe the point information on weekends are carried over from the Friday of the same week with empty number in Vol. We removed these points since they are not consistent with the general pattern and can be misleading in identifying the long-term price fluctuation. We then split the dataset into two groups, using the time stamp November 30, 2022 (the release date of chatGPT) as the cutoff.

```
library(zoo)
```

```
Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':
```

```
as.Date, as.Date.numeric
```

```

library(lubridate)
# [Link: Stack Overflow https://stackoverflow.com/questions/18755643/time-series-analysis-wi

# Create a zoo object
log_return_ts <- zoo(data$log_return[-1], order.by = data$Date[-1])

#remove NA
log_return_ts <- na.omit(log_return_ts)

# Plot the time series
#plot(log_return_ts, main = "Daily Time Series", xlab = "Date", ylab = "Log_return", type = '

# part 3.3
x_train1 <- log_return_ts[index(log_return_ts) <= as.Date("2022-11-30")]
x_test1 <- log_return_ts[index(log_return_ts) > as.Date("2022-11-30")]

# part 3.4
#2321*0.2 # 464 2024-05-21
x_train2 <- log_return_ts[index(log_return_ts) <= as.Date("2024-05-21")]
x_test2 <- log_return_ts[index(log_return_ts) > as.Date("2024-05-21")]

```

Table 1: Summary Statistics of Stock Price

Mean Price	Standard deviation	Minimum	Maximum
69.91683	33.95005	24.4	152.07

Table 1 summarized the descriptive statistics for the stock price from January 3, 2017, to March 27, 2026, including the mean, standard deviation, and range. A more intuitive historical price trajectory is illustrated in Figure 1, where the vertical red dashed line marks the official release of ChatGPT. Overall, the stock shows a sustained upward trend throughout the observed period.

The autocorrelation function (ACF) plot in Figure 2 demonstrates a strong correlation between data points, indicating this time series is non-stationary and does not exhibit white noise characteristics. The ACF values persist well beyond the significance boundaries and decay at a very slow rate. This high correlation is explainable since the stock prices are accumulated based on the previous day's close price. Therefore, data transformation is required to achieve stationarity prior to modeling.

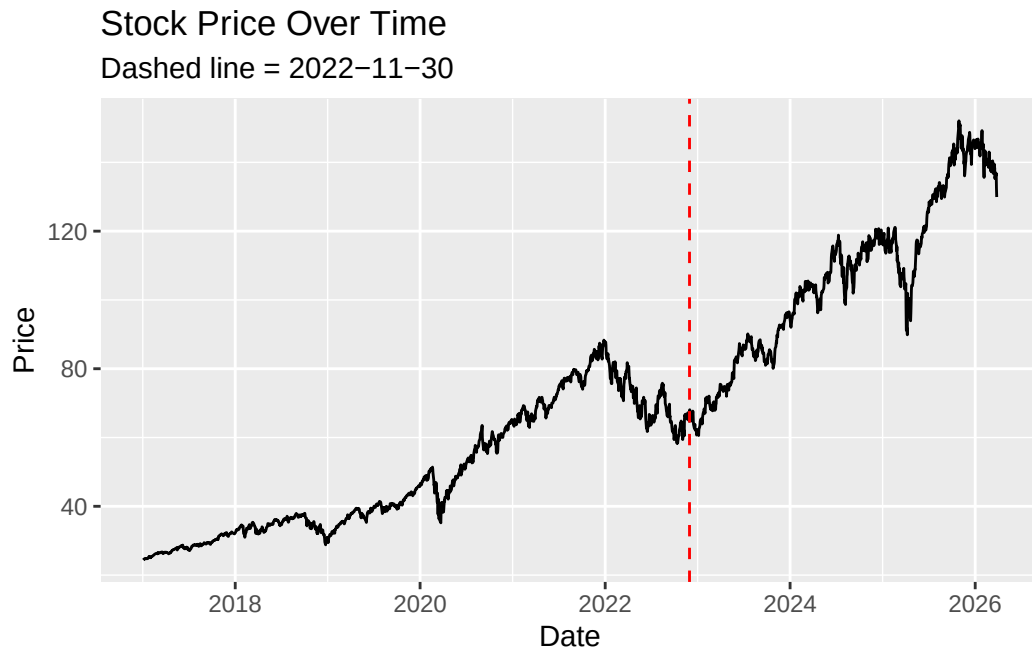


Figure 1: Raw Stock Price Over Time

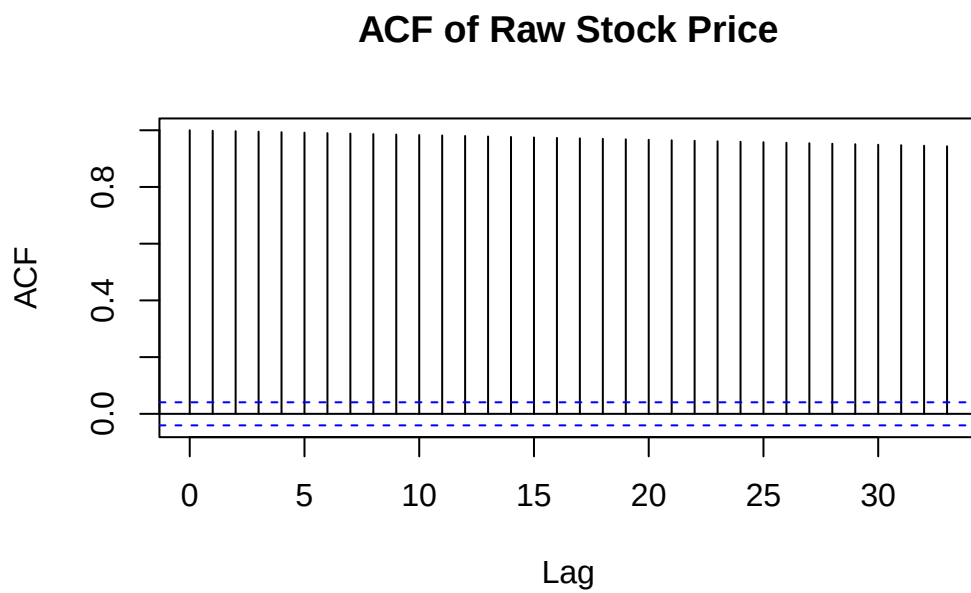


Figure 2: Autocorrelation Function (ACF) of Raw Stock Price.

3 Methods and Results

3.1 Fitting a regression model on price

We fit a regression model with a dummy variable D_t as the predictor.

$$\widehat{Price} = \hat{\beta}_0 + \hat{\beta}_1 t + \hat{\beta}_2 D_t + \hat{\beta}_3 t D_t$$

where $D_t = 0$ represents the period before the appearance of ChatGPT, and $D_t = 1$ represents the period after.

As shown in Figure 3, the two regression lines display different slopes across the two periods. The slope of price over time is steeper after the appearance of AI, meaning the price increased faster during this period.

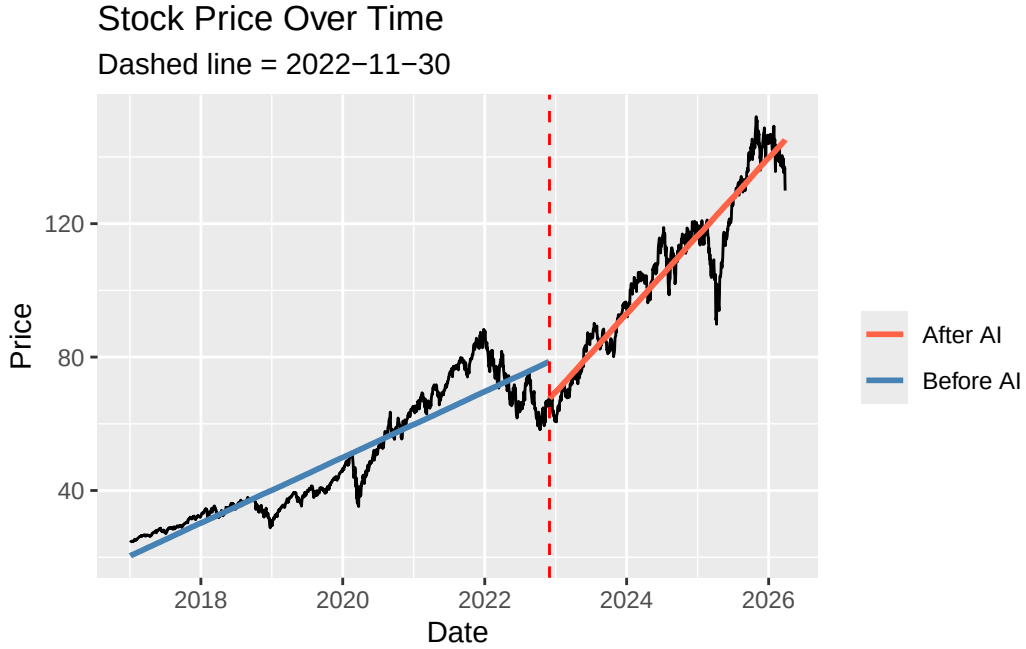


Figure 3: Fitted regression lines for price over time, separated by pre and post-AI periods.

However, the assumptions on regression were not satisfied. The price data is non-stationary, and the QQ plot of the residuals indicates a clear deviation from normality. The Augmented Dickey-Fuller (ADF) test on the model residuals yielded a p-value of 0.0575, failing to reject the null hypothesis of a unit root using 0.01 as the significance level. This suggests that the residuals remain non-stationary.

A robust regression approach was considered too. However, since robust methods do not resolve issues of non-stationarity, this approach was not considered in further analysis. As the residuals are not stationary, an ARMA model cannot be appropriately fitted, and thus no further modelling was performed with regression.

3.2 Differencing on log price

To stabilize the variance and ensure stationarity, we applied a natural logarithmic transformation. The first difference of the log price series, known as the log return, was then used to remove the trend and produce a stationary series suitable for model estimation. Figure 4 shows the log return over time. While the log return mostly fluctuates around a mean of zero with small fluctuations, there are few large spikes. The biggest fluctuations occur around 2020 when the pandemic began, suggesting that the pandemic can cause substantial uncertainty and sharp movements in the market.

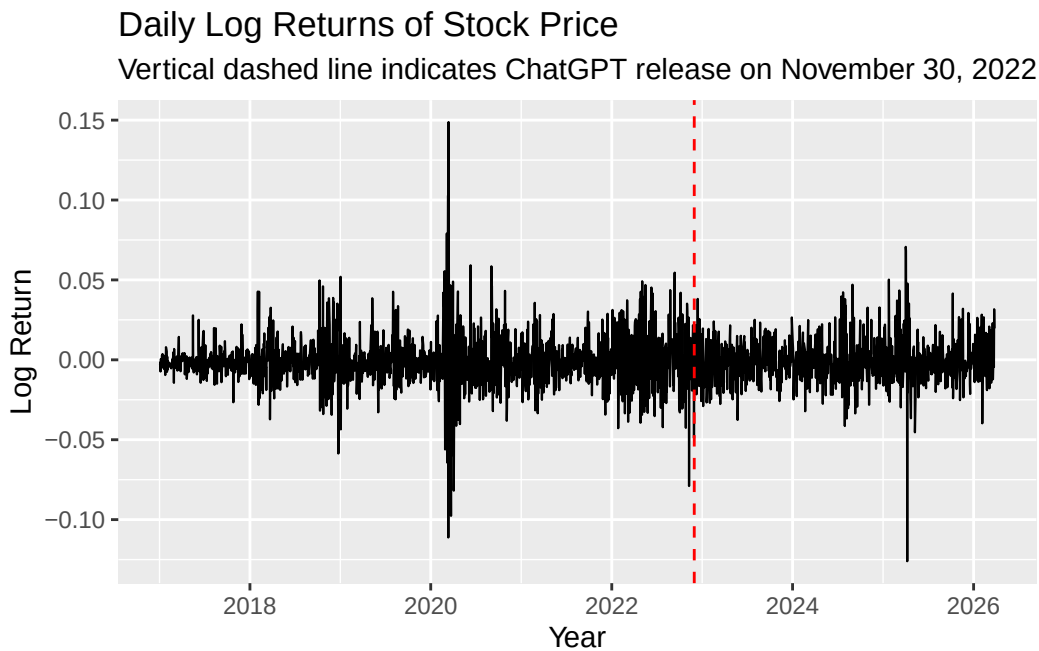


Figure 4: Daily Log Returns of Stock Price (2017–2026).

Although, in Figure 5, the ACF shows there still exists few autocorrelation coefficients outside the significance boundaries, the ADF test was conducted for a formal assessment. The test yielded a p-value of 0.01, providing strong empirical evidence to reject the null hypothesis of a unit root and confirming that the log return series is stationary.

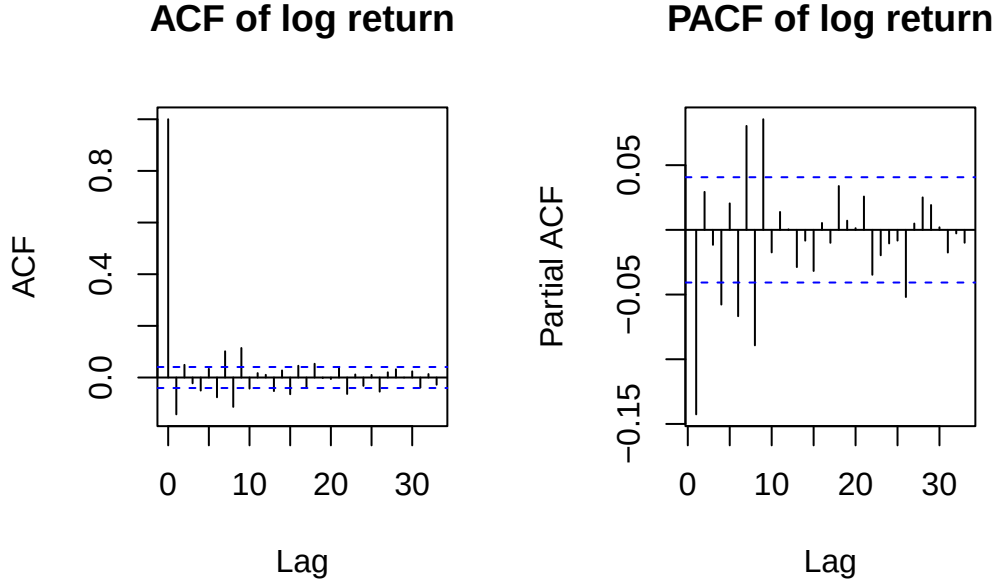


Figure 5: Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of Log Return.

3.3 Using price before AI to predict price after AI

We were interested in how significant the data after November 30, 2022, when ChatGPT got launched, is for forecasting future stock market prices. To do this, we consider data frames of 3 different time sets: January 3, 2017 to November 30, 2022, November 30, 2022 to March 27, 2026, and January 3, 2017 to March 27, 2026.

We aimed to examine whether AI has had an unexpected impact on the stock market and contributed to its fluctuations. We use data from January 3, 2017 to November 30, 2022 to forecast data after November 30, 2022.

We will use seirs from January 3, 2017 to November 30, 2022 to determine an ARIMA model that best explains the data before AI starts. We will use both the AIC, AICc and BIC, and point comparison metrics like MAE, RMSE to determine the best model for forecasting.

We then apply three “benchmark” forecasting methods: the mean method, the naïve method, and the seasonal naïve method. Mean method forecasts all future values as equal to the average (or mean) of the training set. Naïve method set all forecast values to be the value of the last observation of the historical data. Seasonal naïve method set each forecast to be equal to the last observed value from the same season. In this case, we tested for weekly seasonality using this method. The drift method, which accounts for the amount of change over time, and the

decomposition method, which captures the underlying trend, are both considered unnecessary as differencing has already been applied to the log return data.

Lastly, we will compare the performance of ARIMA methods and the benchmark methods.

3.3.1 Fitting ARIMA models

Using the ACF and PACF plots, we can determine the appropriate order of the ARIMA model. According to Figure 5, the ACF gradually tails off, suggesting the presence of autoregressive behavior. According to the PACF, there are strong spikes at lags 2, 4, 7, and 8, and several other significant lags. Based on these patterns, ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(4,1,4), ARIMA(6,1,6), ARIMA(7,1,7), and ARIMA(8,1,8) models were fitted to the log price data to capture the time series.

Table 2: Comparison of Model Performance

	AIC	AICc	BIC	Log Likelihood
ARIMA(1,1,1)	-5.417434	-5.417429	-5.406733	4030.863
ARIMA(2,1,2)	-5.463286	-5.463268	-5.445450	4066.953
ARIMA(4,1,4)	-5.467582	-5.467517	-5.435477	4074.147
ARIMA(6,1,6)	-5.466568	-5.466425	-5.420193	4077.393
ARIMA(7,1,7)	-5.471382	-5.471190	-5.417873	4082.973
ARIMA(8,1,8)	-5.468732	-5.468483	-5.408088	4083.002

winning: 414

```
part_results |> kable(digits=5)
```

	MAE	RMSE	MASE	MASSE
ARIMA(1,1,1)	0.01048	0.01456	0.64101	0.57560
ARIMA(2,1,2)	0.01048	0.01457	0.64107	0.57565
ARIMA(4,1,4)	0.01049	0.01458	0.64138	0.57631
ARIMA(6,1,6)	0.01053	0.01464	0.64430	0.57874
ARIMA(7,1,7)	0.01049	0.01459	0.64166	0.57652
ARIMA(8,1,8)	0.01069	0.01480	0.65359	0.58475

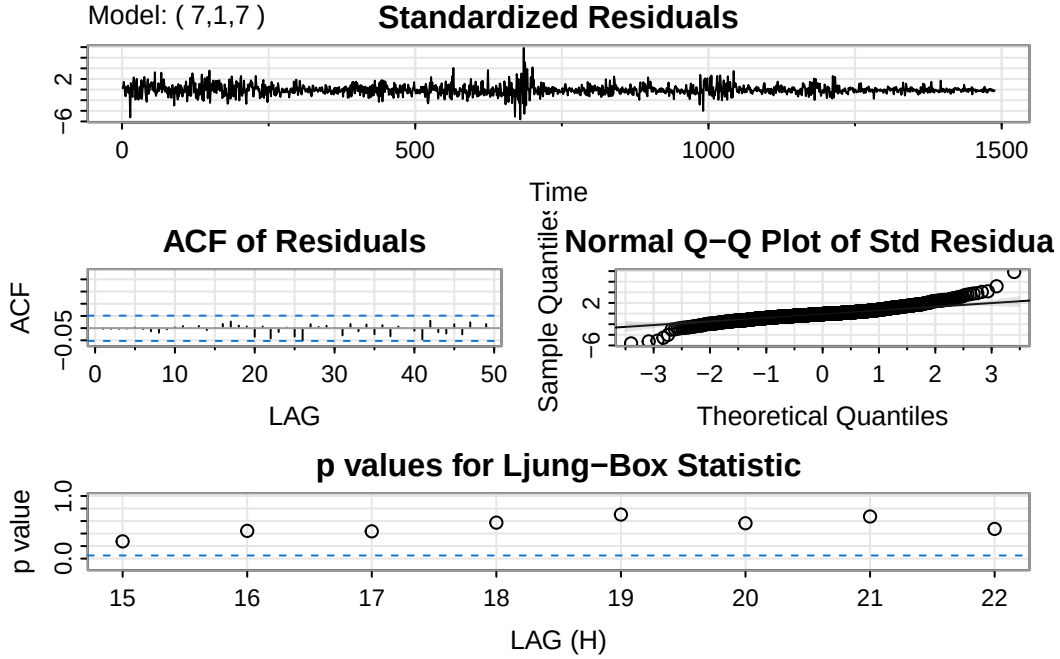


Figure 6: Diagnostic Plot for ARIMA(7,1,7)

According to Table ?? MAE (Mean Absolute Error) and RMSE (Root Mean Squared Error), which measure the difference between actual and predicted values, are small for the mean method but also for all these benchmark methods. This suggests that the data prior to November 30, 2022 can effectively predict the data following the cutoff point. Furthermore, this suggests that AI have not yet contribute a significant impact in the stock market, at least in terms of predictability using these simple benchmark models. As a result, we will use the entire time frame, allocating 80% for training and 20% for testing, to build a forecast model for predicting prices in 2026.

3.3.1 Model Diagnosis

The p values for Ljung-Box Statistic plots indicate that ARIMA(1,1,1) and ARIMA(2,1,2) are not good fits, because p values are insignificant for all lags. ACF plots for ARIMA (4,1,4), ARIMA(7,1,7), and ARIMA (8,1,8) looks stationary, but QQ plots show the points are deviated from the reference line, meaning the normality condition is not met. To determine which model performs the best, we compare the model AIC, AICc, and BIC values.

3.4 Full time frame

Using the ACF and PACF plots, we can determine the appropriate order of the ARIMA model. According to Figure 5, the ACF gradually tails off, suggesting the presence of autoregressive behavior. According to the PACF, there are strong spikes at lags 2, 4, 7, and 8, and several other significant lags. Based on these patterns, ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(4,1,4), ARIMA(7,1,7), and ARIMA(8,1,8) models were fitted to the log price data to capture the time series.

3.3.2 Model Choice

Table 4: Comparison of Model Performance

	AIC	AICc	BIC	logLik
ARIMA(1,1,1)	-5.477657	-5.477655	-5.470223	6357.082
ARIMA(2,1,2)	-5.492856	-5.492849	-5.480465	6376.713
ARIMA(4,1,4)	-5.503769	-5.503742	-5.481465	6393.372
ARIMA(7,1,7)	-5.503271	-5.503192	-5.466098	6398.794
ARIMA(8,1,8)	-5.501308	-5.501206	-5.459179	6398.517

```
full_results |> kable(digits=5)
```

	MAE	RMSE	MASE	MASSE
ARIMA(1,1,1)	0.01127	0.01629	0.71759	0.68425
ARIMA(2,1,2)	0.01127	0.01629	0.71756	0.68423
ARIMA(4,1,4)	0.01127	0.01629	0.71765	0.68418
ARIMA(7,1,7)	0.01127	0.01629	0.71729	0.68425
ARIMA(8,1,8)	0.01127	0.01629	0.71728	0.68426

Among the candidate models, ARIMA(4,1,4) was selected as the best fitting model, as it achieved the lowest AIC, AICc, and BIC values. While ARIMA(7,1,7) produced a very similar AIC, its notably higher BIC indicates that the additional parameters are not justified, making ARIMA(4,1,4) the preferred specification.

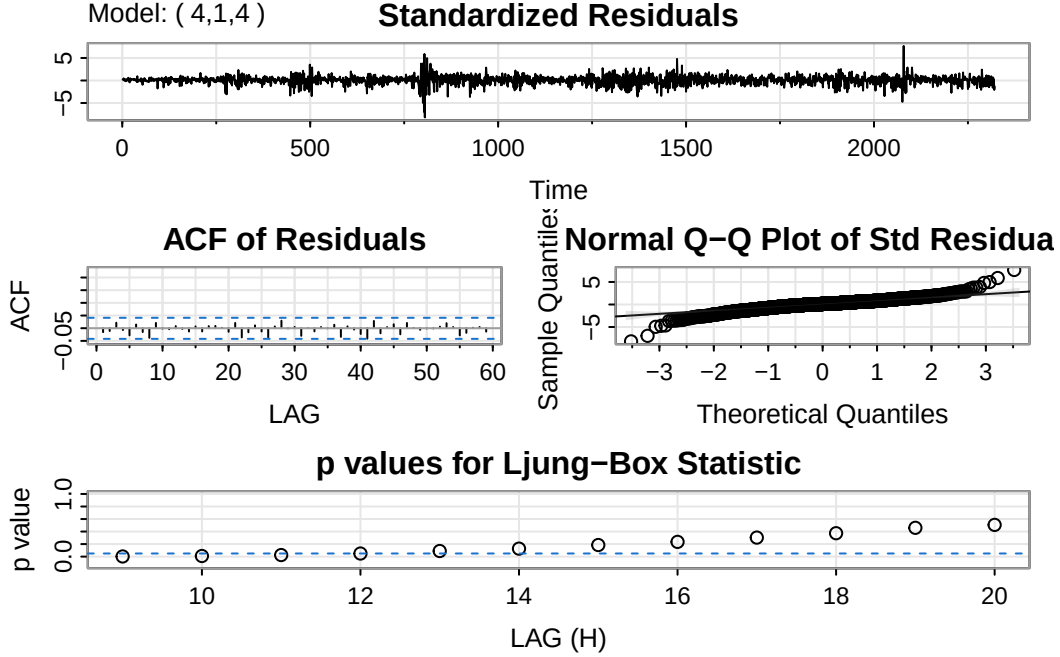


Figure 7: Diagnostic Plot for ARIMA(4,1,4)

Table ?? shows the naive method has the lowest MAE while the mean method has the lowest RMSE. MAPE (Mean Absolute Percentage Error) is a percentage-based error metric that scales forecast errors relative to the actual values. However, MAPE (Mean Absolute Percentage Error) is not included in this comparison because the log return data fluctuates around zero, where some small values closer to zero make MAPE enormous and not a good parameter to look at. MASE (Mean Absolute Scaled Error) and MASSE (Root Mean Squared Scaled Error) are scaled error metrics, and they work by scaling errors based on the training MAE from a simple forecast method. This makes them stable because the denominator is a fixed constant derived from the training data, providing a more informative alternative when MAPE fails to give meaningful results. Based on the lowest MASE and RMSSE values, the mean method performs best overall.

To evaluate whether ARIMA(4,1,4) outperforms the mean method, we split the full time frame into 80% training and 20% testing, fitted ARIMA(4,1,4) on the training set, and computed forecast accuracy metrics as shown in `?@tbl-arima_forecast_check`. Comparing with Table ??, the MASE values are close across both models, however ARIMA(4,1,4) achieves lower MAE, RMSE, and RMSSE, suggesting it is the better performing model.

Next, we construct a prediction interval for the mean method forecast model and ARIMA(4,1,4). In Figure 8, while the histogram suggests the residuals are approximately normally distributed, the QQ-plot indicates that the log returns deviate from normality, violating classic assumptions. Despite this, Figure 9 shows that the prediction

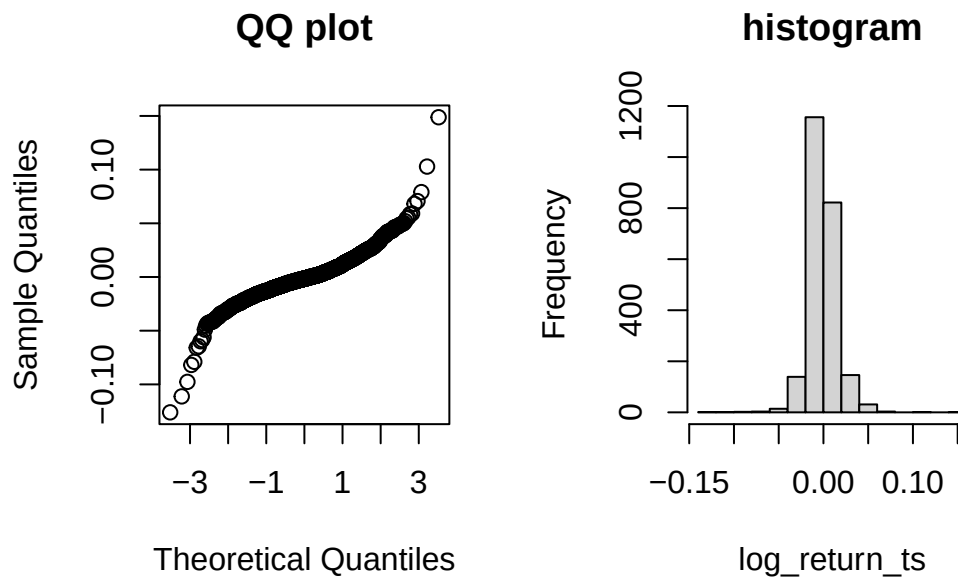


Figure 8: QQ plot and Histogram for log returns

interval successfully captures all observations in the test set, suggesting the interval is functioning as intended. However, the interval is extremely wide, reflecting high uncertainty in the forecasts. This indicates that while the mean method provides valid coverage, it offers little practical predictive power for long-term returns.

Check if ARIMA model would produce a better prediction interval.

```
library(forecast)
```

Attaching package: 'forecast'

The following object is masked from 'package:astsa':

gas

```
# convert keeping dates
fit414_Arima <- Arima(as.numeric(x_train2), order=c(4,1,4), include.mean=FALSE)
fc414 <- forecast(fit414_Arima, h=length(x_test2))
```

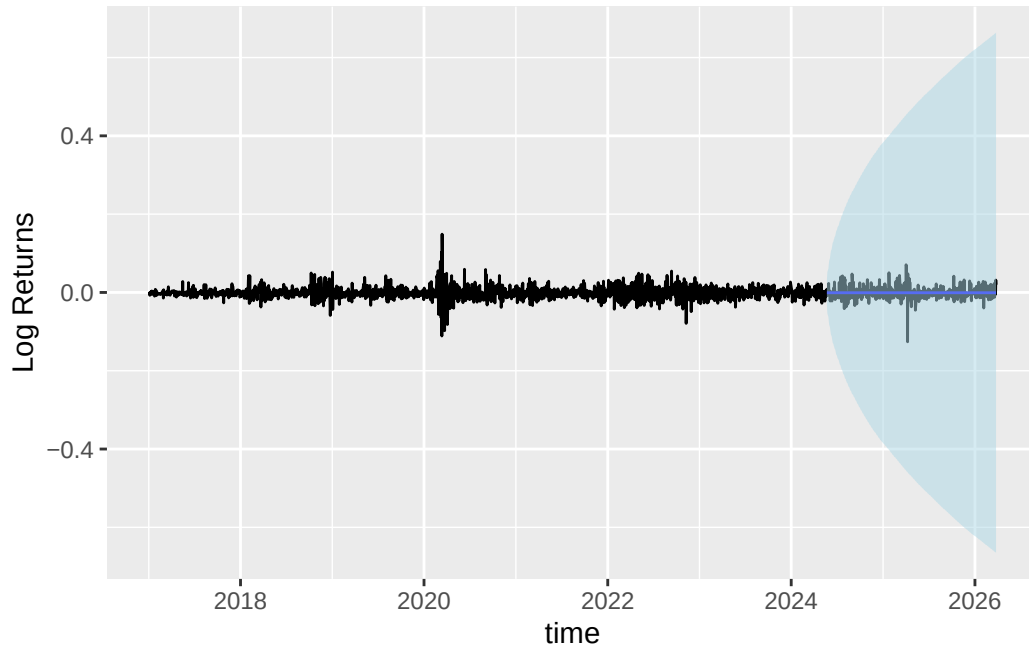
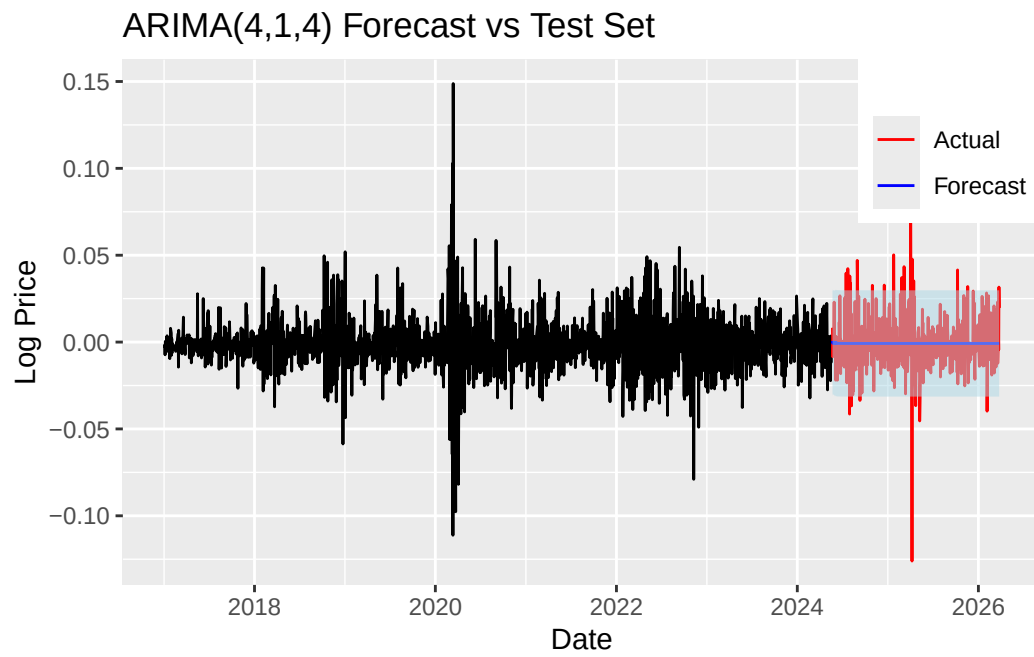


Figure 9: Mean method forecasts and prediction intervals for log returns

```
# build data frames with actual dates
df_train <- data.frame(date=index(x_train2), value=as.numeric(x_train2))
df_test  <- data.frame(date=index(x_test2),  value=as.numeric(x_test2))
df_fc    <- data.frame(
  date  = index(x_test2),
  mean  = as.numeric(fc414$mean),
  lo95  = as.numeric(fc414$lower[,2]),
  hi95  = as.numeric(fc414$upper[,2])
)

ggplot() +
  geom_line(data=df_train, aes(x=date, y=value), color="black") +
  geom_line(data=df_test, aes(x=date, y=value, color="Actual")) +
  geom_line(data=df_fc, aes(x=date, y=mean, color="Forecast")) +
  geom_ribbon(data=df_fc, aes(x=date, ymin=lo95, ymax=hi95), fill="lightblue", alpha=0.5) +
  scale_color_manual(values=c("Actual"="red", "Forecast"="blue")) +
  ggtitle("ARIMA(4,1,4) Forecast vs Test Set") +
  labs(title="ARIMA(4,1,4) Forecast vs Test Set", x="Date", y="Log Price", color="") + theme
```



3.5 Forecast future 2026 data

Using $\text{ARIMA}(4,1,4)$, we forecast the next 30 days starting March 28, 2026. The predicted values appear relatively flat, which is expected for ARIMA models over short period of time. Since future white noise terms are unknown, the forecasts tend to settle toward a constant, resulting in a smoother and less stochastic pattern compared to the real data, as shown in Figure [10](#).

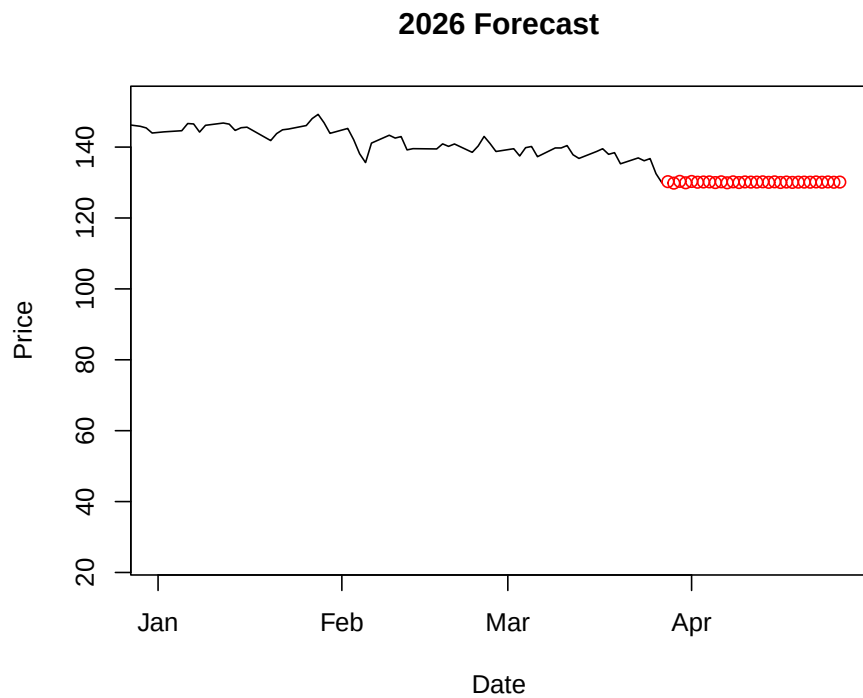
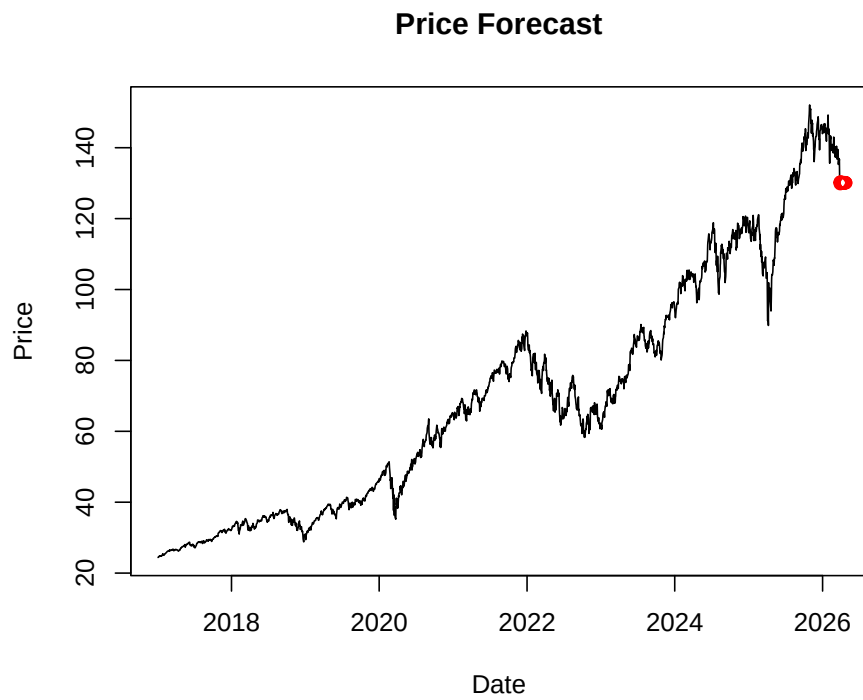


Figure 10: ARIMA(4,1,4) fitted values and forecasts from 2017 to 2026, with a closer view of the 2026 forecast period

4 Conclusions

This project tests whether the rise of AI has significantly affected the behavior of technology sector stock using the XLK ETF price data from 2019 to 2026. Using regression, we compared the price over time before and after the appearance of ChatGPT. It suggests that price after AI increased more quickly than before. However, since price data are non stationary, this result should be interpreted with caution. To address this issue, we transform the data into log return to make the data stationary. Based on this transformation, we find no significant evidence that AI has had a meaningful effect on stock returns. An ARIMA(4,1,4) model is then fitted to capture the time series dynamics.

The second part of this project is to compare different model's predictive power. ARIMA(4,1,4) has the better predictive power compared to benchmark methods with lower MAE, RMSE, and RMSSE. The model is subsequently used to generate forecasts for the next month.

5 Limitations

There are some potential limitations for the dataset. One of the limitations is that there's possibility for State Street to rebalance the stock composition over time, which means the dataset may not remain fully consistent throughout the study period. The companies in the XLK ETF are selected based on the GICS. However, the GICS structure is reviewed annually, meaning State Street need to adjust the XLK portfolio at least on a yearly base. This could reduce the reliability of comparing stock performance across different years and weaken the robustness of analysis.

Another limitation is the dependency of the dataset. In the data analysis, we noticed few large spikes in the log return plot, suggesting factors like pandemics may also attributes to the stock price fluctuation. In Section 3, we found out the time series data points remain highly correlated even after the data transformation. For instance, the QQ plot indicates deviations from normality, suggesting that model assumptions may not hold and prediction intervals may be unreliable. In addition, our model focuses on the short-term impacts after the release of ChatGPT, but the effect of AI may be better reflected in a long term trend rather than a boom. The log transformation may mask the underlying long term trend, because it removes the trend to make the data stationary. The log transformation only focuses on daily changes rather than long-term structural changes. As a results, model based on the log transformation is better at studying short-term market reaction but may be less effective at capturing long-term changes in the market over time.

Also, AR, MA, ARMA are all models that are only a function of the time series itself. They do not include other important factors, such as government policy or economic policy changes. There are many other factors that affects the stock market that we did not incorporate, such as economic indicators and interest rates. Since stock market is influenced by many

external variables, excluding these factors may limit the strength and forecasting accuracy of the models.

6 Future Directions

To account for external factors that might affect the stock market, we could add these as predictors in our model. This can help us better estimate stock prices.

Although we have not been able to conclude AI has significant impact on the behavior of technology sector stock returns, it is still useful to estimate how sensitive technology sector stock price comparing to the broader market indices or other sectors. We might expect the technology sector to show higher volatility and respond more to growth expectations. In future studies, we can fit and compare separate models for different market segments, such as the S&P 500 and individual sectors like healthcare or financials, to better understand these differences.

It is also possible to incorporate the method proposed by Campos-Martins (2024) with our model. After identifying notable development by AI companies, we can adjust our predictors to improve accuracy. For example, we could include an indicator variable that captures major AI events or achievements. This would allow the model to account for changes over time driven by new innovations, instead of relying only on fixed predictors.

Finally, our research should follow a consistent process where we constantly refit the models as the stock market changes over time. Updating the model with new data can help us see whether there are any long-term effects or clear shifts in the data, especially as AI continues to develop and become more widely used.